EE4-47
EE9-CS6-1

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2021

MSc and EEE PART IV: MEng and ACGI

MODELLING AND CONTROL OF MULTI-BODY MECHANICAL SYSTEMS

Friday, 30 April 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions

All questions carry equal marks.

This is an OPEN BOOK examination.

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : S.A. Evangelou
Second Marker(s) : A. Astolfi

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MODELLING AND CONTROL OF MULTIBODY MECHANICAL SYSTEMS

1. A pendulum system consists of a massless beam of length l fixed to Earth at point O and attached at each of its two ends with separate pendulums, as shown in Figure 1.1. The pendulums have mass m_1 and m_2 suspended from massless rods of length l_1 and l_2, respectively. The beam and the masses m_1, m_2 are free to swing in a vertical plane respectively by an angle θ with respect to Earth, and θ_1 and θ_2 with respect to the beam, under the influence of gravity; note that the angles θ_1 and θ_2 are measured with respect to the direction perpendicular to the beam.

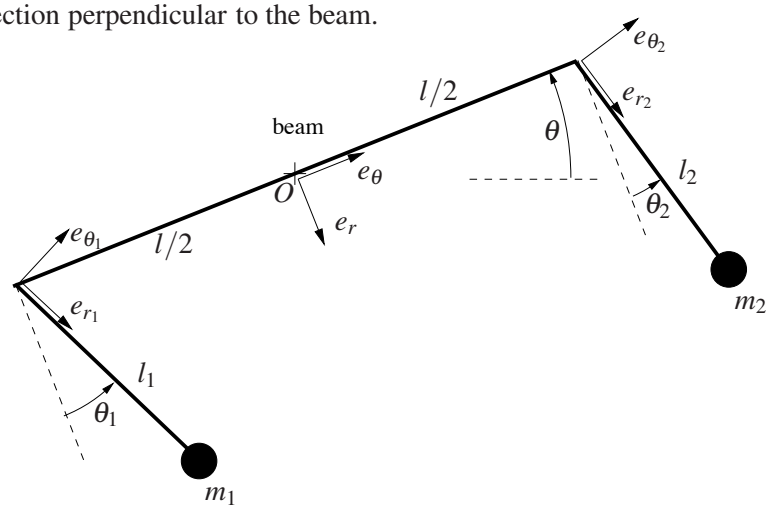


Figure 1.1 Pendulum system with a beam and two pendulums.

Three moving Cartesian coordinate systems with unit vectors $\boldsymbol{e}_r$, $\boldsymbol{e}_\theta$ for the first, $\boldsymbol{e}_{r_1}$, $\boldsymbol{e}_{\theta_1}$ for the second, and $\boldsymbol{e}_{r_2}$, $\boldsymbol{e}_{\theta_2}$ for the third, attached respectively to the beam (at point O) and the two pendulums (at their respective joint to the two ends of the beam), are used to analyse the motion of the two masses of the pendulum system.

a) Write

   i) the position vector, and [2]

   ii) the velocity vector [4]

   of each of the masses, in the inertial reference frame.

b) Compute

   i) the kinetic energy, [2]

   ii) the potential energy, and [3]

   iii) the Lagrangian function [1]

   of the system.

c) Derive the Lagrangian equations of motion of the pendulum system for

   i) the θ_1 generalised coordinate, and [3]

   ii) the θ_2 generalised coordinate. [3]

d) Write the Lagrangian equation of motion for the θ generalised coordinate but do not expand it. [2]

2. A quadrotor drone set up for a lab environment is attached to a smooth horizontal rail at point O that restricts its degrees of freedom to two, as shown in Figure 2.1. Thus, the drone is free to rotate in a vertical plane about a normal-to-the-diagram axis through O by an angle θ, and translate along the rail (at point O) by a distance x, under the influence of gravity and a drone-generated control force and torque. The centre of mass of the drone is at point C at distance a below O. The mass and spin inertia (about its centre of mass) of the drone are m and I respectively. The control force, F_{th}, delivered by equal thrust from each of the propellers, acts on the drone always in the same direction with respect to the drone, as shown in Figure 2.1. The control torque, T_{th}, (not shown) delivered by differential thrust from the propellers acts on the drone about a normal to the diagram axis.

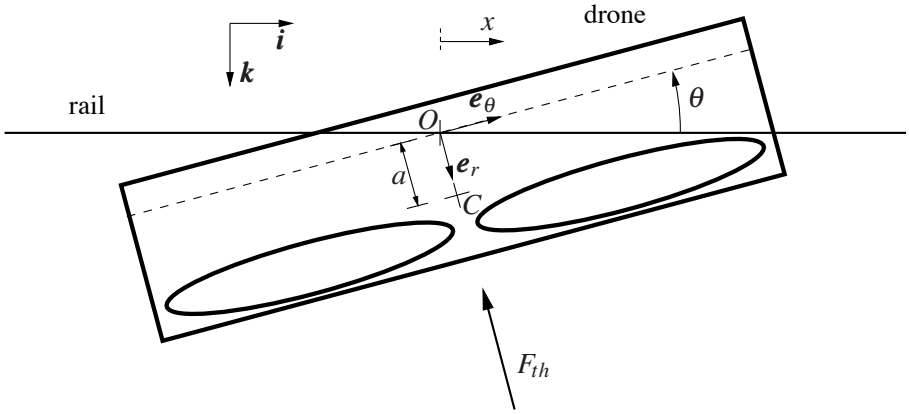


Figure 2.1 Two-degree of freedom quadrotor drone.

An Earth-fixed Cartesian coordinate system with unit vectors $\boldsymbol{i}$ and $\boldsymbol{k}$, and a moving Cartesian coordinate system with unit vectors $\boldsymbol{e_r}$ and $\boldsymbol{e_\theta}$ are used to analyse the motion of the drone. The moving coordinate system is attached to the drone, it has its origin at O and it rotates by an angle θ.

a) Write the position vector of the centre of mass of the drone. [3]

b) Calculate the velocity vector of the centre of mass of the drone. [3]

c) Compute the total kinetic energy of the drone. [3]

d) Calculate the total potential energy of the drone, and hence determine the Lagrangian function. [3]

e) Use the Lagrangian approach to derive the equations of motion of the drone in terms of the generalised coordinates. [8]

3. As in Q2, a quadrotor drone set up for a lab environment is attached to a smooth horizontal rail at point O that restricts its degrees of freedom to two, as shown in Figure 3.1. Thus, the drone is free to rotate in a vertical plane about a normal-to-the-diagram axis through O by an angle θ, and translate along the rail (at point O) by a distance x, under the influence of gravity and a drone-generated control force and torque. The centre of mass of the drone is at point C at distance a below O. The mass and spin inertia (about its centre of mass) of the drone are m and I respectively. The control force, F_{th}, delivered by equal thrust from each of the propellers, acts on the drone always in the same direction with respect to the drone, as shown in Figure 3.1. The control torque, T_{th}, (not shown) delivered by differential thrust from the propellers acts on the drone about a normal to the diagram axis.

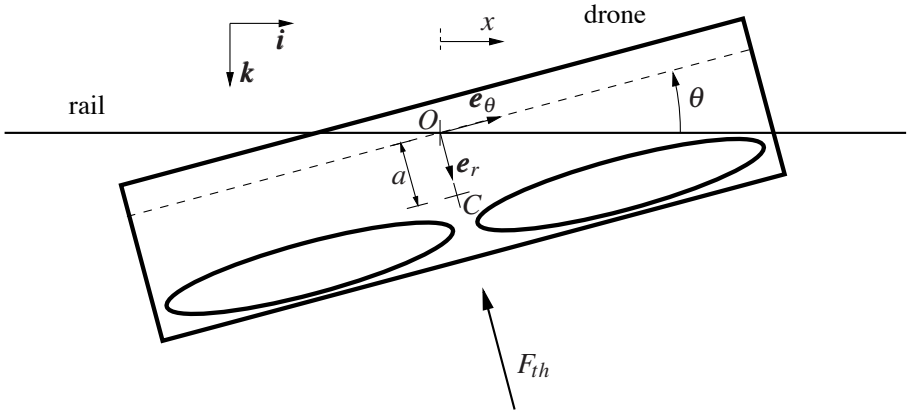


Figure 3.1 Two-degree of freedom quadrotor drone.

An Earth-fixed Cartesian coordinate system with unit vectors $\boldsymbol{i}$ and $\boldsymbol{k}$, and a moving Cartesian coordinate system with unit vectors $\boldsymbol{e_r}$ and $\boldsymbol{e_\theta}$ are used to analyse the motion of the drone. The moving coordinate system is attached to the drone, it has its origin at O and it rotates by an angle θ.

a) Determine the generalised coordinates of the system. [2]

b) Write the acceleration vector of the centre of mass of the drone. [3]

c) F_v is the unknown constraint force on the drone from the rail in the $\boldsymbol{k}$ direction. Write the total force that acts on the drone in vector form. [3]

d) Use the vectorial approach to derive F_v in terms of the generalised coordinates and the control force, F_{th}. [5]

e) Use the vectorial approach to further derive the equations of motion of the drone with respect to the generalised coordinates, and the control force and torque. [7]

4. As in Q1, a pendulum system consists of a massless beam of length l fixed to Earth at point O and attached at each of its two ends with separate pendulums, as shown in Figure 4.1. The pendulums have mass m_1 and m_2 suspended from massless rods of length l_1 and l_2, respectively. The beam and the masses m_1, m_2 are free to swing in a vertical plane respectively by an angle θ with respect to Earth, and θ_1 and θ_2 with respect to the beam, under the influence of gravity; note that the angles θ_1 and θ_2 are measured with respect to the direction perpendicular to the beam.

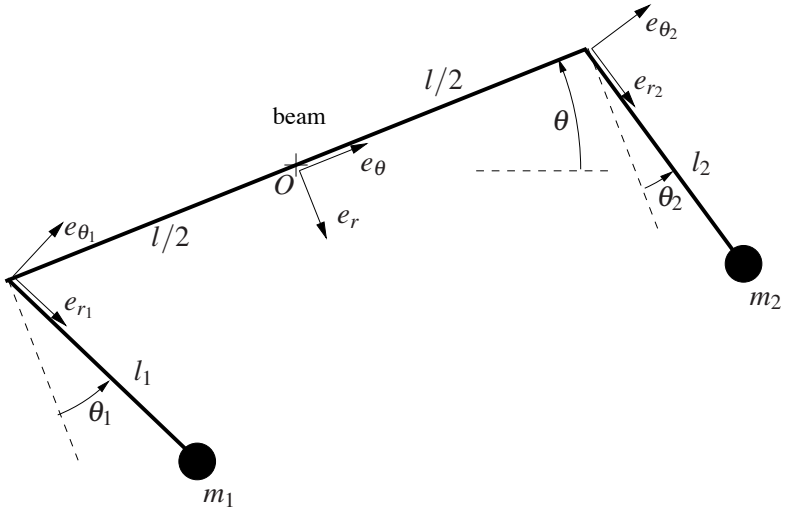


Figure 4.1 Pendulum system with a beam and two pendulums.

Three moving Cartesian coordinate systems with unit vectors $\boldsymbol{e}_r$, $\boldsymbol{e}_\theta$ for the first, $\boldsymbol{e}_{r_1}$, $\boldsymbol{e}_{\theta_1}$ for the second, and $\boldsymbol{e}_{r_2}$, $\boldsymbol{e}_{\theta_2}$ for the third, attached respectively to the beam (at point O) and the two pendulums (at their respective joint to the two ends of the beam), are used to analyse the motion of the two masses of the pendulum system.

a) Write the number of degrees of freedom and the generalised coordinates of the system. [2]

b) Write each of $\boldsymbol{e}_r$ and $\boldsymbol{e}_\theta$ in terms of the basis vectors $\boldsymbol{e}_{r_1}$ and $\boldsymbol{e}_{\theta_1}$. Also write each of $\boldsymbol{e}_r$ and $\boldsymbol{e}_\theta$ in terms of the basis vectors $\boldsymbol{e}_{r_2}$ and $\boldsymbol{e}_{\theta_2}$. [2]

c) Calculate the acceleration vector of the mass m_1 in the inertial reference frame in terms of $\boldsymbol{e}_{r_1}$ and $\boldsymbol{e}_{\theta_1}$. [3]

d) Hence, use the <u>vectorial</u> approach to

   i) derive one of the equations of motion of the pendulum system in terms of the generalised coordinates, and [3]

   ii) compute the force of constraint in the pendulum rod holding mass m_1 in terms of the generalised coordinates. [3]

e) Calculate the acceleration vector of mass m_2 in the inertial reference frame in terms of $\boldsymbol{e}_{r_2}$ and $\boldsymbol{e}_{\theta_2}$. [3]

f) Hence, use the <u>vectorial</u> approach to

   i) derive a second equation of motion of the pendulum system in terms of the generalised coordinates, and [2]

   ii) compute the force of constraint in the pendulum rod holding mass m_2 in terms of the generalised coordinates. [2]